On Range Ambiguities From Elevation Grating Lobes in Ship Detection RCM SAR Imagery

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***Abstract*— As a 2-D spatial filter, the unidirectional electronically steerable planar phased-array synthetic aperture radar (SAR) antenna of the radar constellation mission (RCM) samples the space–time process in azimuth and elevation, respectively, with spatial intervals and visible region defined by the corresponding interelement spacings and wavenumbers. Since position and wavenumber are conjugate Fourier transform variables, interelement spacings need to abide by the Nyquist–Shannon criterion. Furthermore, since the antenna's response is a 2π-periodic function, it produces an infinite number of periodically spaced peaks that are equal in strength to the main lobe peak. If any of these peaks is shifted to the visible region, due to azimuth and/or elevation steering, it becomes a grating lobe that warrants attention. This article investigates grating lobe ambiguities due to elevation steering (E-GLAs) in ScanSAR RCM imagery. E-GLAs resemble legitimate targets and they pose a challenging source of false alarms (FAs) for operational ship detection. Using theoretical analysis, simulations, and a real-world Ship Detection dataset, it is demonstrated that E-GLAs become visible for off-nadir angles $\geq$50.38°. A novel algorithm is presented to automatically locate E-GLAs in repeat-pass images. Efficacy of the algorithm is demonstrated by tracking a verified E-GLA in a stack of 26 × 2 RCM images near Galápagos Islands, processed using downlinked and definitive satellite orbits, respectively. Results based on definitive orbits are shown to outperform their downlinked counterparts.**

***Index Terms*— Elevation steering, false alarm (FA), grating lobe, radar constellation mission (RCM), range ambiguity, repeat pass, ScanSAR, ship detection, steep incident angle, synthetic aperture radar (SAR).**

I. Introduction

SINCE real-world data made available to the research community by Seasat, one of the first spaceborne synthetic aperture radars (SARs), ambiguities have been recognized as being a major source for false alarms (FAs) [1]. Despite the significant advancements achieved in the SAR field, today ambiguities continue to pose a challenge not only to the ongoing SAR missions but also to the newly envisioned ones [2], [3], [4], [5]. This article investigates one of the most challenging FAs from an operational ship detection perspective. That is, elevation grating lobe ambiguities (E-GLAs). E-GLAs are difficult to recognize because they resemble legitimate targets and they can fool top performing constant FA rate (CFAR) detectors. In fact, this study was triggered by the operators of Fisheries and Oceans Canada (DFO) observing what at a first glance appeared to be a dark vessel (DV) near Galápagos Islands. In partnership with Ecuador, DFO has deployed state-of-the-art DV detection (DVD) program to help protect the recently expanded Galápagos Islands Marine Reserve, an area regarded as one of the most unique and biodiverse marine ecosystems on the planet, from the threats posed by industrial-scale foreign fishing fleets [6], [7], [8]. For performing ship detection in radar constellation mission (RCM) imagery, the trimodal (3MD) based CFAR detector is operationally deployed[1] [9], [10]. As it will be expounded throughout this article, the supposed DV turned out to be an E-GLA.

The IEEE Standard for Definitions of Terms for Antennas defines a grating lobe as, "a lobe other than the main lobe, produced by an array antenna when the interelement spacing is sufficiently large to permit the in-phase addition of radiated fields in more than one direction" [11]. In signal processing, aliasing occurs when a continuous real-valued bandlimited signal is undersampled using a sampling rate that does not abide by the Nyquist–Shannon criterion. That is, $f_s \leq 2B$, where B is the largest frequency present in the signal. Similarly, from a spatial sampling perspective, the antenna is a 2-D spatial filter and the radar return signal will be weighted by the corresponding antenna pattern and can exhibit grating lobes based on the azimuth and elevation spacings between the subarrays and the corresponding azimuth and elevation steering angles, respectively.

Grating lobe ambiguities have been previously reported in the literature for the extended high (EH) Mode of both RADARSAT-1 and RADARSAT-2 [12], [13]. At high incidence angles the main lobe of the antenna is sufficiently far away from the physical boresight that a significant grating lobe on the opposite side of the physical boresight manifests, with a gain proportional to the steering angle. Particularly, using the experimental EH4 imaging mode of RADARSAT-1 with horizontal (HH) polarization, E-GLA was demonstrated where the main lobe of the antenna was directed at the ocean off the Newfoundland coast with a high incidence angle of 55°, and the grating lobe pointing at Lake Champlain on the Vermont/New York border. Since there is a small main lobe return from the ocean at high incidence angles, the grating lobe

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[1]For the 3MD-based CFAR detections reported in this article, probability of FA (PFA) = 1×10^{-10}.

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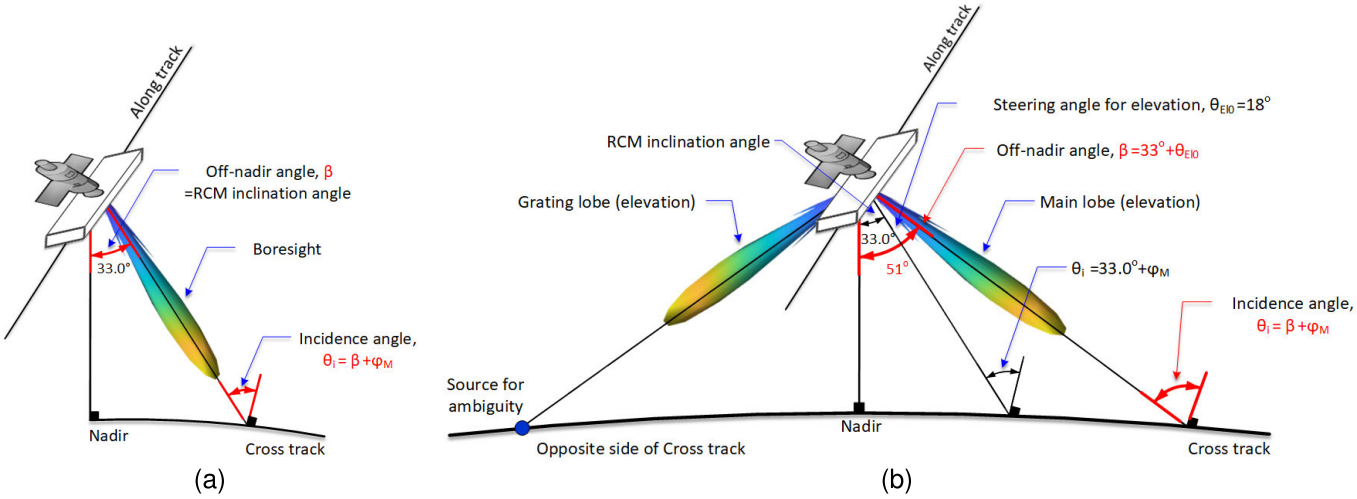


Fig. 1. Acquisition geometry. (a) No elevation steering. (b) Elevation steering with $\theta_{\mathrm{El}_0} = 18°$ (see Appendix A).

return was found to be dominant [12]. Similar observations are reported for RADARSAT-2 where it is stated that HH beams 4, 5, and 6 of the EH mode are designed for imaging near the North and South poles and are not recommended for other regions because E-GLA can appear at incidence angles above 54° and become progressively more severe with increasing incidence angle [13].

A similar phenomenon is also observed when squinting the SAR antenna in the along-track direction. That is, grating lobe ambiguity due to azimuth steering (A-GLA). For example, in the wide-swath terrain observation by progressive scan (TOPS) imaging mode with TerraSAR-X (TSX), the antenna is steered along-track during the acquisition of a burst. It is noted that A-GLA arises when the pointing angle is steered away from boresight, and for increasing steering angle the grating lobe gain increases while the main lobe gain decreases [14]. In contrast, researchers tried to harness the azimuth grating lobe effect by presenting a bidirectional imaging mode that provides for simultaneous imaging of two directions in the along-track direction by one antenna into one receiving channel. Results were demonstrated for a short-term time series of images and interferograms acquired by the TSX and TanDEM-X satellites [15].

This article investigates E-GLAs in Ship Detection RCM SAR imagery. To reduce FAs in real-world operational settings, a novel algorithm is introduced to automatically locate an E-GLA in secondary repeat-pass acquisitions given a main image. A major challenge handled by the algorithm is that the pixel location in the secondary repeat-pass acquisitions relative to the main image shifts either toward or away from the radar, proportional to perturbations in the relative satellite state vectors. The remainder of this article is organized as follows. Section II introduces relevant details on grating lobes. To demonstrate the effect of elevation steering on grating lobes, Section III provides simulations for the planar phased-array RCM SAR antenna. The repeat-pass RCM SAR dataset utilized in this study is described in Section IV. A preliminary analysis for a 3MD-based CFAR detection of an E-GLA ambiguity tracked in the repeat-pass dataset is discussed in Section V. The proposed algorithm is presented in Section VI. Results and Discussion are given in Section VII. This article is concluded in Section VIII.

TABLE I
SIMULATION PARAMETERS FOR THE RCM SAR ANTENNA

Parameter	Value
Center frequency	5.405 GHz
Bandwidth	100 MHz
Aperture length×height	6.75×1.37 m^2
Number of radiating elements	4608
Element type	Dual polarization patch radiators
Number of elements per sub-array	24
Number of subarrays per column	32
Number of columns per panel	3
Number of panels	2
Element spacing	0.8445λ (0.0469 m)
Row spacing	0.77λ (0.0427 m)
Azimuth beamwdith*	0.202°
Elevation beamwidth*	3.43°

* Azimuth and elevation beamwidths are chosen after RADARSAT-2 and they are for full antenna active, uniform phase distribution [16]. The simplified simulations presented in this paper are meant to visually demonstrate the elevation grating lobes and they are not fully representative of the real-world RCM SAR antenna.

II. GRATING LOBES

The active phased-array antenna of RCM SAR has a total of $M \times N$ radiating elements in the along-track (i.e., azimuth) and cross-track (i.e., elevation) directions, respectively (see Table I). The antenna lies in the yz-plane (see Figs. 1–3, and Appendix A for the relationship between off-nadir β and incidence θ_i angles) where its electric field in the far-field (or Fraunhofer) region is given by [17]

$$E_F(\theta_{\mathrm{El}}, \phi_{\mathrm{Az}}) = E_0(\theta_{\mathrm{El}}, \phi_{\mathrm{Az}})\mathrm{AF}(\theta_{\mathrm{El}}, \phi_{\mathrm{Az}}) \quad (1)$$

here, $E_0(\theta_{\mathrm{El}}, \phi_{\mathrm{Az}})$ is the electric field pattern of the element, AF is the array factor, θ_{El} is the elevation angle, and ϕ_{Az} is the azimuth angle. AF is defined by [18]

$$\mathrm{AF}(\theta_{\mathrm{El}}, \phi_{\mathrm{Az}}) = \mathbf{w}^{\mathbf{H}}\mathbf{s}(\theta_{\mathrm{El}}, \phi_{\mathrm{Az}}) \quad (2)$$

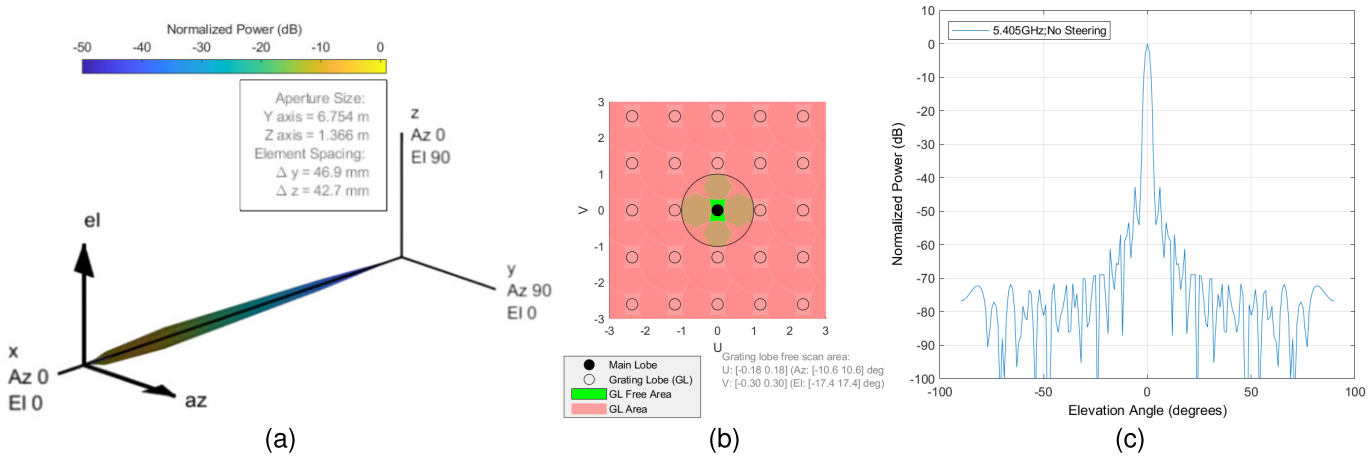


Fig. 2. Simulation results for the RCM antenna without elevation steering. (a) 3-D response pattern. (b) Grating lobe diagram in uv space. (c) Cartesian plot for elevation cut θ_{El}, $\phi_{\mathrm{Az}} = 0°$.

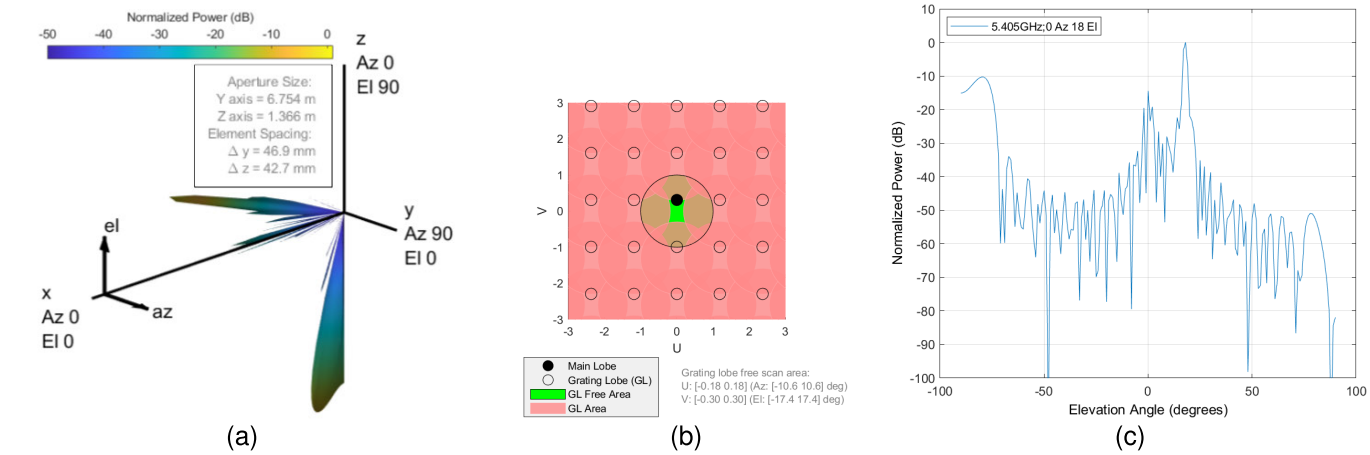


Fig. 3. Simulation results for the RCM antenna with elevation steering $\theta_{\mathrm{El}_0} = 18°$. (a) 3-D response pattern. (b) Grating lobe diagram in uv space. (c) Cartesian plot for elevation cut θ_{El}, $\phi_{\mathrm{Az}} = 0°$.

where $\mathbf{s}(\theta_{\mathrm{El}}, \phi_{\mathrm{Az}})$ is the steering vector for directions of arrival, and the weight vector $\mathbf{w}$ is chosen to maximize the response of the array for a particular arrival direction. The directions of maximum spatial response are determined by the peaks of AF. To avoid spatial aliasing (i.e., grating lobes) for a signal with a wavenumber k, the signal must be sampled at spatial intervals

$$d < \frac{\pi}{k_{\max}} = \frac{\lambda_{\min}}{2} \tag{3}$$

where $k_{\max}$ is the largest wavenumber present in the signal [19]. AF depends on the wavenumber components of the wavefield in the array plane k_y and k_z, respectively, which are related to the look direction of the arriving field by [20]

$$k_y = -\frac{2\pi}{\lambda}\sin(\phi_{\mathrm{Az}})\cos(\theta_{\mathrm{El}}), \quad k_z = -\frac{2\pi}{\lambda}\sin(\theta_{\mathrm{El}}) \tag{4}$$

where the look direction points away from the array to the wavefield.[2] Since AF is the discrete time Fourier transform (DTFT) of E_0, and because DTFT is 2π-periodic, it produces an infinite number of periodically spaced peaks that are equal in strength to the main lobe peak. If the array steered to ϕ_{Az_0} and θ_{El_0}, the main lobe peak would be at k_{y0} and k_{z0}, and for integer values $m = 0, \pm1, \pm2, \ldots,$ and $n = 0, \pm1, \pm2, \ldots,$ respectively, the array would also have strong peaks at

$$k_{ym} = k_{y0} + m\frac{2\pi}{d_y}, \quad k_{zn} = k_{z0} + n\frac{2\pi}{d_z} \tag{5}$$

where d_y and d_z are the interelement spacings in the azimuth and elevation directions, respectively. Because grating lobe diagrams are provided in Section III in the uv coordinate system, it is useful to represent grating lobes in terms of directional cosines as follows:

$$u_m = u_0 - m\frac{\lambda}{d_y}, \quad v_n = v_0 - n\frac{\lambda}{d_z}. \tag{6}$$

In terms of the look direction, one can compute a look direction $(\phi_{\mathrm{Az}_{m,n}}, \theta_{\mathrm{El}_{m,n}})$ using [20]

$$u_m = \sin\left(\phi_{\mathrm{Az}_m}\right)\cos\left(\theta_{\mathrm{El}_m}\right), \quad v_n = \sin\left(\theta_{\mathrm{El}_n}\right). \tag{7}$$

The grating lobe is visible for $u_m^2 + v_n^2 \le 1$, and it is nonvisible for $u_m^2 + v_n^2 > 1$.

[2]See [21] for more details on the spherical coordinate system deployed in the Phased-Array System Toolbox™ of MATLAB and how to convert between the various coordinate systems. This MATLAB toolbox is used for simulating the RCM SAR antenna discussed in this article.

III. Simulating RCM SAR Antenna's Response

This simulation is carried out to illustrate the effect of elevation steering on grating lobes for the RCM antenna. The simulation parameters in Table I are chosen to closely mimic those of the planar distributed phased-array RCM antenna. For the real-world RCM antenna, the long dimension (6.75 m) of the array is directed along the direction of flight, while the short direction (1.37 m) is oriented such that the normal to the physical antenna aperture is inclined toward the Earth's surface at an angle of 33° with respect to the vertical between the spacecraft and the ground. Hence, the spacecraft has 33° roll. It supports dynamic azimuth yaw steering for Earth rotation compensation, 0° at the poles and 3.9° at the equator. The HH spacing between subarray elements is $d_y = 0.8445\lambda$ (0.0469 m). The vertical spacing between rows is $d_z = 0.77\lambda$ (0.0427 m). These interelement spacings were carefully chosen to balance various tradeoffs to fulfill the overall steering requirements for the mission [16], [22], [23], [24], [25]. To simplify the simulations, the inclination angle for the antenna is fixed at 0°. As shown in Appendix A, because the RCM antenna is inclined at an angle of 33° and with incidence angles for the RCM mission $\theta_i \in [17.3°, 58.71°]$ [22], the required elevation steering for the RCM antenna would be

$$\theta_{\mathrm{El}_0} = \theta_i - 33° - \varphi_M \tag{8}$$

where φ_M is the core angle between the nadir point and the grazing point measured at the Earth's center.

A criterion for determining the maximum element spacing $d_{z,\max}$ in the elevation direction for an array steered to angle θ_{El} is to set the spacing so that the nearest grating lobe is at the horizon. From (4) and (5)

$$2\pi \frac{d_z}{\lambda}\left(\sin(\theta_{\mathrm{El}}) - \sin\left(\theta_{\mathrm{El}_0}\right)\right) = 2\pi n, \quad n = 0, \pm 1, \pm 2, \ldots \tag{9}$$

Hence, elevation grating lobes occur at angles

$$\sin\left(\theta_{\mathrm{El}_n}\right) = \sin\left(\theta_{\mathrm{El}_0}\right) + n\frac{\lambda}{d_z}, \quad n = 0, \pm 1, \pm 2, \ldots \tag{10}$$

And for $n = \pm 1$ [19], [26]

$$d_{z,\max} \leq \frac{\lambda}{1 \pm \sin\left(\theta_{\mathrm{El}_0}\right)}. \tag{11}$$

Fig. 4 plots $d_{z,\max}$ from (11) as a function of the elevation steering angles θ_{El_0} for the RCM mission. The interelement elevation spacing $d_{z,\mathrm{RCM}}$ for the RCM antenna is overlaid on the plot. While the plot shows that elevation steering for the RCM mission in the negative direction is not impacted by grating lobes, it is depicted that grating lobes would affect steering with elevation angles $\theta_{\mathrm{El}_0} \geq 17.38°$. This corresponds to radar off-nadir (or look) angles $\beta \geq 50.38°$.

The acquisition geometry for elevation steering of $\theta_{\mathrm{El}_0} = 0°$ and $\theta_{\mathrm{El}_0} = 18°$, respectively, are shown in Fig. 1(a) and (b). Representative simulation results for elevation steering of $\theta_{\mathrm{El}_0} = 0°$ (i.e., no steering) and $\theta_{\mathrm{El}_0} = 18°$, respectively, are presented in Figs. 2 and 3. These elevation angles would correspond to real-world radar incidence angles θ_i of 36.5° and 58.1°, respectively [27]. Simulation results are presented in terms of, (a) 3-D radiation pattern, (b) grating lobe diagram, and (c) elevation pattern in rectangular coordinates, respectively.

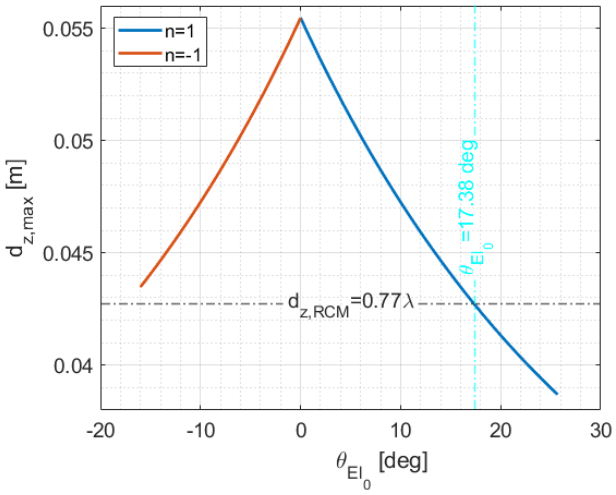


Fig. 4. $d_{z,\max}$ versus elevation steering angles θ_{El_0} for RCM.

The grating lobe diagram displays positions of the peaks of the narrowband array pattern, only near the location of the main lobe in the range $[-3, 3]$ in uv coordinates. Visible and nonvisible grating lobes are displayed as open circles, and the main lobe is indicated as a filled black circle. The visible region for which $u^2 + v^2 \leq 1$ is shown as a unit circle centered at the origin. The grating lobe-free region, shown in green, is the range of the main lobe for which there are no grating lobes in the visible region [28].

For elevation steering angle $\theta_{\mathrm{El}_0} = 0°$ in Fig. 2(b) the main lobe is inside the green region and there is no grating lobes within the visible region. However, as depicted in Fig. 3(b), for $\theta_{\mathrm{El}_0} = 18°$, because the main lobe is outside the green area, there is a grating lobe within the visible region. Appearance of the grating lobe can be further verified by inspecting the 3-D response patterns and the rectangular plots in Figs. 2 and 3, respectively. These results demonstrate that the RCM antenna would have a prominent elevation grating lobe at the higher range of the radar incident angles which as depicted in Fig. 4 corresponds to $\theta_{\mathrm{El}_0} \geq 17.38°$ or $\beta \geq 50.38°$, where gain of the grating lobe is proportional to the elevation steering angle.

IV. Repeat-Pass Ship Detection RCM Dataset

As shown in Fig. 5, a total of 26 × 2 repeat-pass ground-range (GRD) Ship Detection (i.e., ScanSAR) scenes from the three satellites RCM-1, RCM-2, and RCM-3 in the vicinity of Galápagos Islands are utilized in this study [29]. Half of these scenes are processed using downlinked orbits, while the second half utilizes the more accurate definitive orbits.[3]

[3]The definitive orbit data are usually available within 24 h from image acquisition. Geolocation accuracy requirements for downlinked orbits are <50 or <100 m, depending on imaging mode. For definitive orbits, the geolocation accuracy requirements are <25 m for Med Res 16 m mode, while it is <10 m for High Res 5 m mode, Very High Res 3 m mode and Spotlight [30]. For this study, in addition to the archived Ship Detection dataset with downlinked orbits, the dataset was reprocessed with definitive orbits.

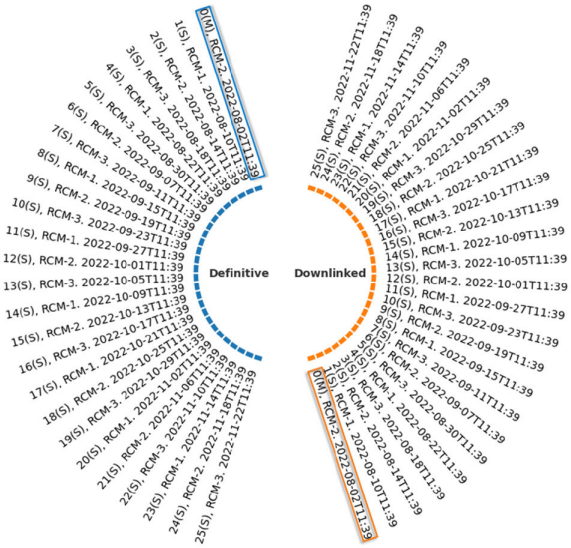


Fig. 5. 26 × 2 repeat-pass Ship Detection RCM scenes utilized in this study. The 1 × 2 acquisitions highlighted by blue and orange rectangles, respectively, are the main images for the definitive and downlinked orbits.

The polarization is HH, and the georeferenced RCM image pixel data files are oriented such that North is nominally up and East is nominally on the right. Because the satellite pass direction is descending and the antenna pointing is right, these RCM scenes are flipped from left to right. The swath width for each scene is 350 km, and it is covered by ten beams named SC-28 to SC-37 which support incidence angles $\in[39.82°, 58.24°]$ for the [near, far] ranges, respectively. The beams have pulse bandwidth that varies from 100 to 25 MHz for beams SC-28 to SC-37, respectively. Hence, the slant-range resolution is fixed per beam but it varies from 1.5 m for beam SC-28 to 6 m for beam SC-37. Furthermore, the GRD resolution depends on the incidence angle and it is degraded from beams SC-37 to SC-28, respectively, by a factor $\in[1.1761, 1.5616]$. The `sampledPixelSpacing` and `sampledLineSpacing` for the GRD product are 4 and 16 m, respectively. Also, `numberOfRangeLooks` = 5 and `numberOfAzimuthLooks` = 1 [27].

While each of these RCM acquisitions has 14 subproducts, in this investigation, we are focusing on subproduct no. 5. As noted in Section I, this dataset is originally acquired for DFO's DVD program in partnership with Ecuador to help protect the recently expanded Galápagos Islands Marine Reserve from the threats posed by industrial-scale foreign fishing fleets [6], [7], [8]. For performing ship detection in RCM imagery, the 3MD-based CFAR detector is operationally deployed [9], [10]. It is noted that one prominent 3MD-based CFAR detection in burst 140 from subproduct no. 5 could not be verified neither through association with automatic identification system (AIS) messages nor via using conventional range ambiguity analysis. Figs. 6 and 7 depict subproduct no. 5 which is comprised of 277 bursts from ten beams. Fig. 8 provides a zoom-in into burst 140, where the above-noted 3MD-based CFAR detection named E-GLA detection (E-GLA-D) is located. To illustrate the strength of E-GLA-D, representative 3-D plots for sigma0 [dB] pertaining to repeat-pass acquisitions of RCM-3 (September 11, 2022), RCM-1 (September 15, 2022), RCM-3 (September 23, 2022), and RCM-1 (September 27, 2022), respectively, are provided in Fig. 9. A list of relevant statistics for E-GLA-D from these four repeat-pass acquisitions is provided in Table II. The plots and representative statistics for E-GLA-D clarify why this detection mimics a legitimate target and hence it was confidently picked up by the 3MD-based CFAR detector.

V. Preliminary Analysis

Since E-GLA-D could neither be associated with any AIS messages nor an originating source for a range ambiguity, we tracked E-GLA-D in 26 × 2 repeat-pass acquisitions. The rationale here is that if E-GLA-D is not a DV but rather a recurrent detection for a fixed structure or a range ambiguity with its source outside the image boundaries, one would expect it to persist at the exact same geographic (longitude, latitude) location in the repeat-pass images. To investigate this hypothesis, the following procedure is followed.

1) Take E-GLA-D from subproduct no. 5 acquired on August 02, 2022 as the main detection.
2) Using the center (pixel, line) coordinates of the main E-GLA-D, perform Forward Geolocation to find the corresponding center geographic coordinates (longitude, latitude) in degrees.
3) Perform Reverse Geolocation to find the corresponding center secondary (estGeoPixel, estGeoLine) coordinates.
4) In each secondary repeat-pass acquisition, manually locate the center (truthGeoPixel, truthGeoLine) coordinates.
5) Calculate the error, (estGeoPixel−truthGeoPixel, estGeoLine−truthGeoLine).
6) Perform this process for the 25 × 2 repeat-pass stack processed using both downlinked and definitive orbits, respectively, acquired from August 10, 2022, to November 22, 2022.

The metadata that comes in the product information file (PIF) of each RCM georeferenced image includes three forms of information that can be used in geolocation: orbit data, tie points and rational functions. In this article, the tie points are used to perform the Forward Geolocation while the rational functions are utilized for the Reverse Geolocation. The tie points are derived using a rigorous projection model (RPM) and they provide a regular grid of paired image and ground coordinates between which the geolocation of intermediate points can be interpolated. The RPM maps how image coordinates project to ground coordinates. The RPM is based on the intersection of the range sphere centered on the satellite position, the zero-Doppler plane passing through the satellite position, and an Earth model. The rational functions are a set of polynomials that provide an accurate approximation of the reverse RPM. Details on the Forward and Reverse Geolocation algorithms can be found in [31], [32], and [33].

By following the above procedure one can accurately predict the center line location in the secondary images. However,

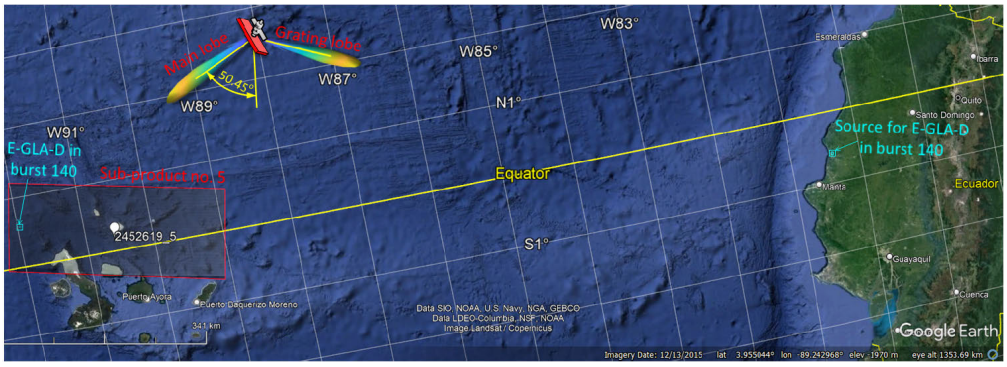


Fig. 6. Geometry and footprint for subproduct no. 5 of the Ship Detection RCM stack off the Galápagos Islands (Credit to Google Earth. Data: SIO, NOAA, U.S. Navy, NGA, GEBCO, LDEO-Columbia, NSF. Image: LANDSAT/Copernicus). The 3MD-based CFAR E-GLA-D detection (i.e., see Fig. 8) is marked by the small cyan rectangle and it is located in the far-left side of the RCM image in burst 140 of beam SC-37 with incidence angle $\theta_i = 57.43°$ and a corresponding off-nadir angle $\beta = 50.45°$. The E-GL points to the opposite cross-track direction where the source of the ambiguity is located and marked with a cyan rectangle on the right side of the image.

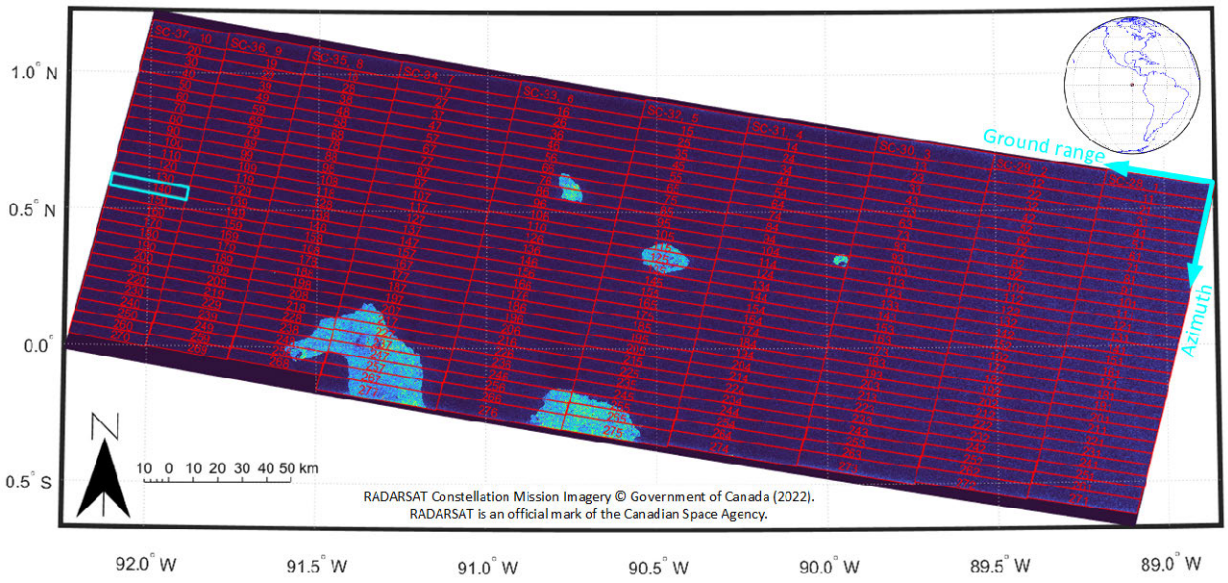


Fig. 7. Subproduct no. 5 for the main RCM-2 image acquired on August 02, 2022 (see Figs. 5 and 6). The image is comprised of 277 bursts from ten beams. Burst 1 is located at the top right corner of the image in the first column which corresponds to the first beam SC-28, while the last column corresponds to the tenth beam SC-37. Highlighted by a cyan rectangle is burst 140 (see Fig. 8) where E-GLA-D is located.

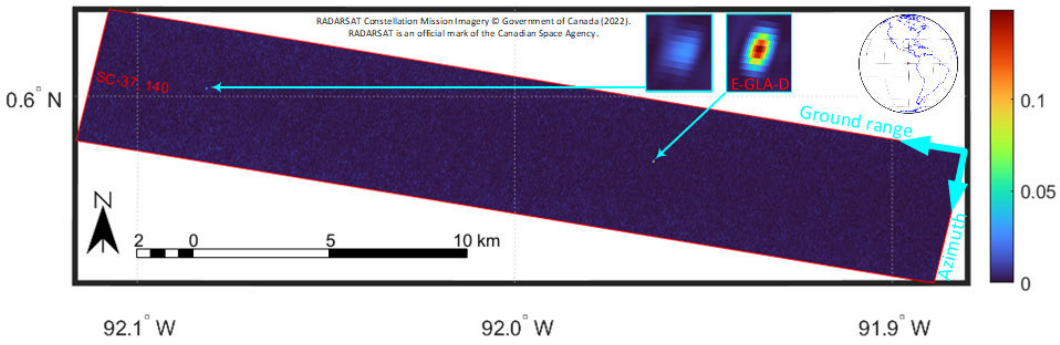


Fig. 8. Main burst 140 of beam SC-37 from subproduct no. 5 in Fig. 7. Two verified E-GLAs are highlighted. Although the strongest ambiguity on the right (E-GLA-D) is detected by 3MD-based CFAR, it is an FA that will be tracked in the subsequent 25 × 2 repeat-pass acquisitions.

as depicted in Fig. 10, predicting the center pixel location in the secondary images is subject to an error that varies unexpectedly from 1 pixel to more than 125 pixels. This variation in the secondary center pixel location suggests that if E-GLA-D is indeed an elevation grating lobe ambiguity, it should be sensitive to variations in the nadir position (i.e., satellite ground

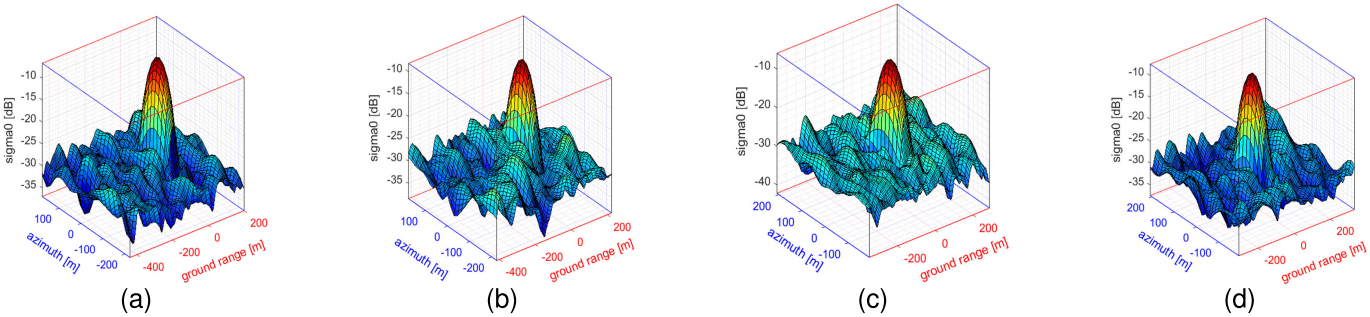


Fig. 9. Sigma0 [dB] plots for 3MD-based CFAR detection of E-GLA-D in four repeat-pass acquisitions. (a) RCM-3, September 11, 2022. (b) RCM-1, September 15, 2022. (c) RCM-3 September 23, 2022. (d) RCM-1, September 27, 2022.

TABLE II
REPRESENTATIVE STATISTICS FROM FOUR REPEAT-PASS ACQUISITIONS FOR E-GLA-D OF FIG. 9

No.	center longitude [deg]	center latitude [deg]	θ_{iS} [deg]	ref noise [dB]	DN Power Stats [dBsm] mean	peak	min	sigma0 Stats [dB] mean	peak	min	std*
07(S), RCM-3	−91.9628	0.578 44	57.425	−31.43	37.10	55.66	25.58	−25.97	−7.41	−37.50	7.62
08(S), RCM-1	−91.9632	0.578 47	57.426	−33.44	37.10	54.96	24.61	−25.98	−8.11	−38.46	7.08
10(S), RCM-3	−91.9640	0.578 52	57.427	−31.30	37.93	56.51	26.02	−25.14	−6.56	−37.05	7.66
11(S), RCM-1	−91.9648	0.578 50	57.429	−31.30	38.32	57.23	20.83	−24.75	−5.84	−42.20	7.88

* One standard deviation away from the mean.

track) of the secondary relative to the main RCM satellite. To investigate this hypothesis the following procedure is carried out for images from both the downlinked and definitive orbits.

1) Take 1 × 2 E-GLA-D from sub-product no. 5 acquired on August 02, 2022 as the main detection with the truth center pixel as the mainPixel.
2) In the 25 × 2 repeat-pass acquisitions dated from August 10, 2022 to November 22, 2022, take the truth center pixel of E-GLA-D as the secondaryPixel.
3) Calculate

$$\text{truthShift} = \text{secondaryPixel} - \text{mainPixel}. \tag{12}$$

4) Interpolate satellite state vectors for each RCM acquisition to the zero-Doppler time of the center (pixel, line) where E-GLA-D is located. The interpolation procedure followed utilizes Hermite polynomials and it is described in [34].
5) Using the interpolated position satellite state vectors calculate

$$\text{satPos}_{M,S} = \sqrt{x_{M,S}^2 + y_{M,S}^2 + z_{M,S}^2} \tag{13}$$

$$\text{satPos}_{\text{Dif}} = \text{satPos}_S - \text{satPos}_M. \tag{14}$$

6) Using the interpolated satellite velocity state vectors calculate

$$\text{satSpeed}_{M,S} = \sqrt{v_{xM,S}^2 + v_{yM,S}^2 + v_{zM,S}^2} \tag{15}$$

$$\text{satSpeed}_{\text{Dif}} = \text{satSpeed}_S - \text{satSpeed}_M. \tag{16}$$

Using (12), (14), and (16), respectively, a stacked plot for the downlinked and definitive orbits is provided in Fig. 11. The plot depicts a strong correlation between truthShift [pixels] and satPosition$_{\text{Dif}}$ [m], with slight shifts between the downlinked and definitive orbits. Also, correlation between

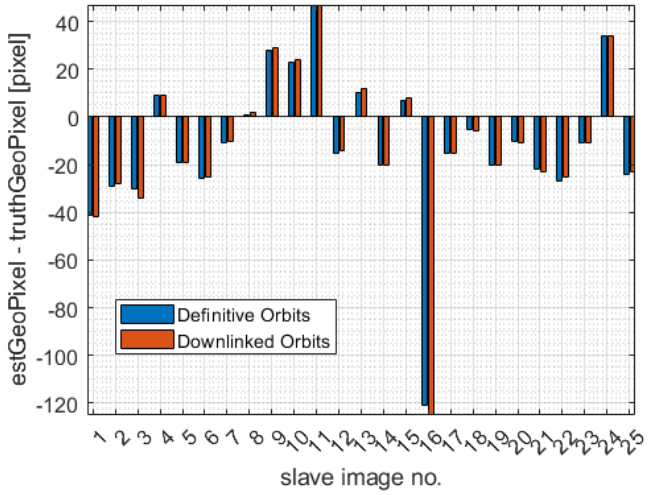


Fig. 10. Secondary pixel error = estGeoPixel−truthGeoPixel [pixels] for E-GLA-D when tracked in 26 × 2 repeat-pass acquisitions, processed using downlinked and definitive orbits, respectively.

truthShift [pixels] and satSpeed$_{\text{Dif}}$ [m/s] is observed, albeit note how satSpeed$_{\text{Dif}}$ based on definitive orbits is more closely correlating with truthShift when compared to downlinked orbits. In the proposed algorithm only the satellite position state vectors are needed.

VI. PROPOSED ALGORITHM

Fig. 12 depicts the geometry of two repeat-pass acquisitions, one is for a main image and the second is for a secondary image. The satellite position and velocity orbit state vectors in earth-centered, earth-fixed (ECEF) coordinate system are given by, (x_M, y_M, z_M), (v_{xM}, v_{yM}, v_{zM}), (x_S, y_S, z_S), and (v_{xS}, v_{yS}, v_{zS}), respectively. The RCM satellites are traveling in a 120 m radius orbital tube [30]. Hence, the orbital distance

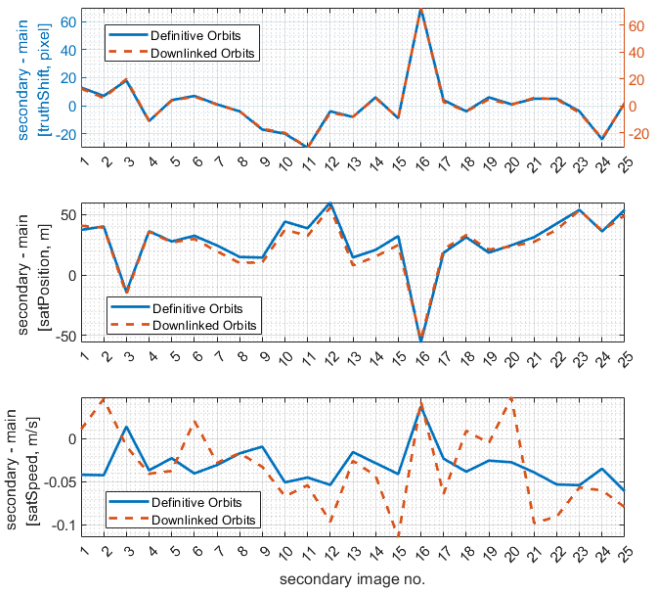


Fig. 11. Stacked correlation plot for truthShift [pixels], satPositionDif [m], and satSpeedDif [m/s] of E-GLA-D in the secondary image relative to the main image. Results are provided in terms of both downlinked and definitive orbits.

between any two repeat-passes can be up to 240 m, which leads to slight variations in the corresponding state vectors. These variations cause the geographic GRD pixel location in the secondary image to shift either toward the radar or away from it, depending on perturbations in the secondary vectors relative to the main ones. This effect has already been demonstrated by the correlations noted in the stacked Fig. 11.

The angles subtended at the center of Earth C by the ground range arcs BN_M and BN_S, respectively, are given by

$$\varphi_M = \frac{R_{\mathrm{GM}}}{R_E}, \quad \varphi_S = \frac{R_{\mathrm{GS}}}{R_E}. \tag{17}$$

Applying the law of cosines on triangle $A_M BC$ for the main acquisition gives

$$R_M^2 = R_E^2 + (R_E + h_M)^2 - 2R_E(R_E + h_M)\cos(\varphi_M). \tag{18}$$

Hence, from (17)

$$R_M^2 = R_E^2 + (R_E + h_M)^2 - 2R_E(R_E + h_M)\cos\left(\frac{R_{\mathrm{GM}}}{R_E}\right). \tag{19}$$

And solving for R_{GM} yields

$$R_{\mathrm{GM}} = R_E \arcsin\left(\frac{R_E^2 + (R_E + h_M)^2 - R_M^2}{2R_E(R_E + h_M)}\right). \tag{20}$$

Similarly, applying the law of cosines on triangle $A_S BC$ gives

$$R_S^2 = R_E^2 + (R_E + h_S)^2 - 2R_E(R_E + h_S)\cos(\varphi_S). \tag{21}$$

Hence, from (17)

$$R_S^2 = R_E^2 + (R_E + h_S)^2 - 2R_E(R_E + h_S)\cos\left(\frac{R_{\mathrm{GS}}}{R_E}\right). \tag{22}$$

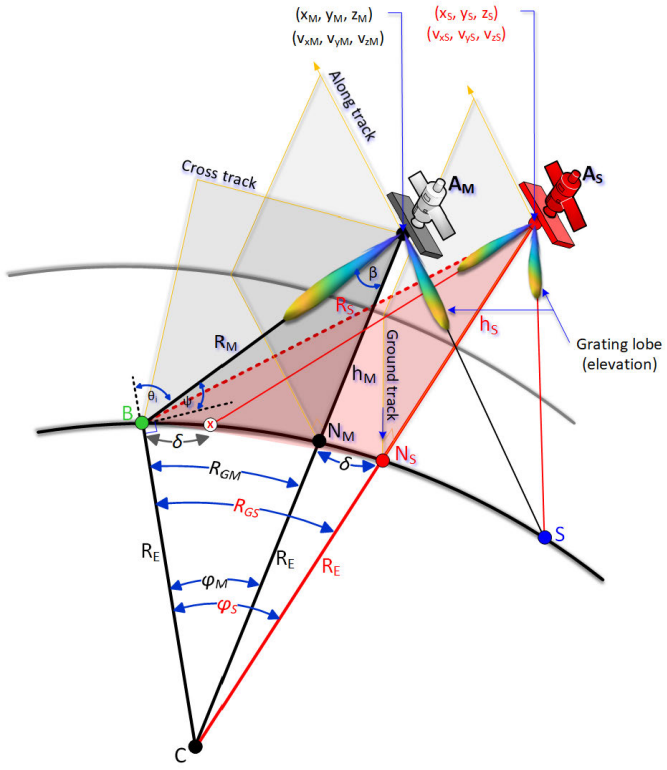


Fig. 12. Geometry of the proposed algorithm for estimating the center pixel of E-GLA-D in a secondary image given the center (pixel, line) in a main image.

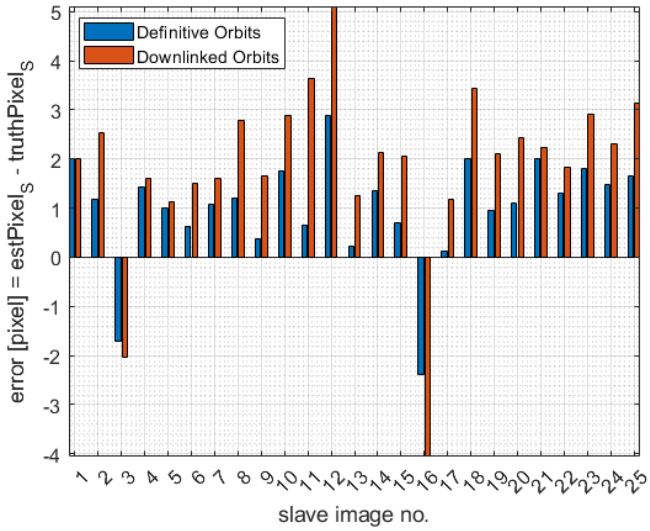


Fig. 13. $\mathrm{errorPixel}_S$ for E-GLA-D tracked using the proposed algorithm in 25 × 2 secondary repeat-pass images with reference to the 1 × 2 main image. Results are provided in terms of both downlinked and definitive orbits.

And solving for R_{GS} yields

$$R_{\mathrm{GS}} = R_E \arcsin\left(\frac{R_E^2 + (R_E + h_S)^2 - R_S^2}{2R_E(R_E + h_S)}\right). \tag{23}$$

Accordingly, the shift distance in meters for the secondary acquisition relative to the main one is defined as follows:

$$\delta = R_{\mathrm{GS}} - R_{\mathrm{GM}}. \tag{24}$$

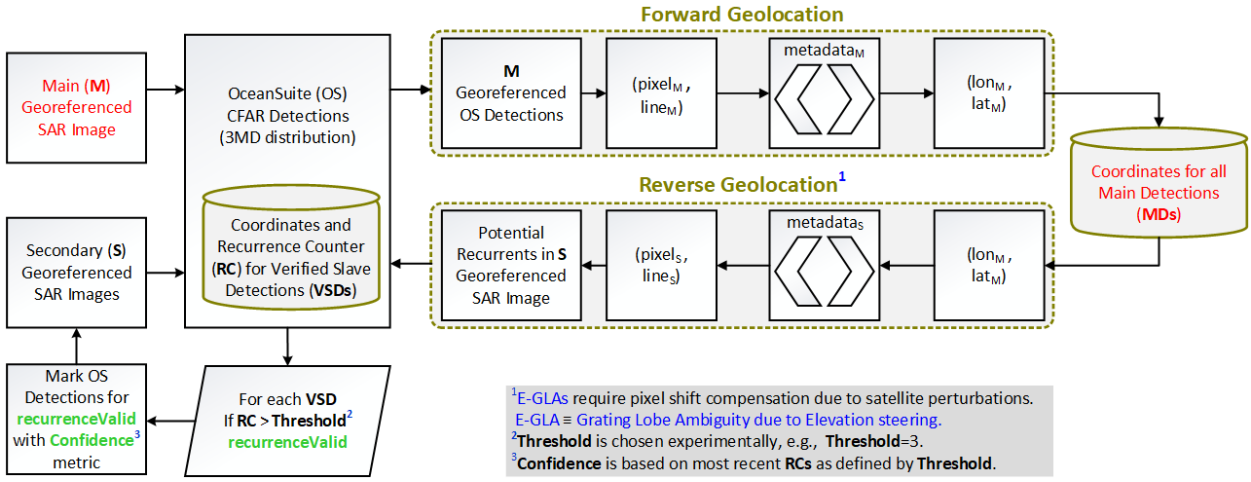


Fig. 14. Proposed strategy for augmenting OceanSuite to identify recurrent detections including E-GLAs in repeat-pass RCM imagery.

See Appendix B for details on computing R_E at geographical point B in Fig. 12. For `sampledPixelSpacing` in meters, the shift in pixels is

$$\delta_p = \frac{\delta}{\texttt{sampledPixelSpacing}}. \tag{25}$$

The above analysis assumes equal `samplesPerLine` for both the main and secondary images. If `samplesPerLine` is not equal then it needs to be compensated for as follows:

$$\Delta = \texttt{samplesPerLine}_S - \texttt{samplesPerLine}_M. \tag{26}$$

And the corrected pixel location in the secondary image relative to the main one would be given by

$$\delta_{pC} = \delta_p - \Delta. \tag{27}$$

A concise summary for the proposed algorithm is provided in Algorithm 1. The algorithm computes a center secondary $(\mathbf{pixel_S}, \mathbf{line_S})$ given a center main $(\mathbf{pixel_M}, \mathbf{line_M})$ for an E-GLA-D. In the algorithm provided, M ≡ main, S ≡ secondary, meta ≡ metadata, SV ≡ State Vector, lon ≡ longitude, lat ≡ latitude, interp2zeroDopTime ≡ interpolate SV to zero-Doppler time, fwdGeoLoc ≡ forward Geolocation, and revGeoLoc ≡ reverse Geolocation.

Algorithm 1 Proposed Algorithm for Computing an E-GLA-D's $(\mathbf{pixel_S}, \mathbf{line_S})$ given $(\mathbf{pixel_M}, \mathbf{line_M})$

MAIN2SECONDARY($\mathbf{pixel_M}, \mathbf{line_M}, \mathbf{meta_M}, \mathbf{meta_S}$)
For input main center $(\mathbf{pixel_M}, \mathbf{line_M})$
$\mathbf{SV_M}$ = interp2zeroDopTime($\mathbf{pixel_M}, \mathbf{line_M}, \mathbf{meta_M}$)
$\mathbf{lon_M}, \mathbf{lat_M}$ = fwdGeoLoc($\mathbf{pixel_M}, \mathbf{line_M}, \mathbf{meta_M}$)
Find secondary center $(\mathbf{pixel'_S}, \mathbf{line_S})$
$\mathbf{pixel'_S}, \mathbf{line_S}$ = revGeoLoc($\mathbf{lon_M}, \mathbf{lat_M}$)
$\mathbf{SV_S}$ = interp2zeroDopTime($\mathbf{pixel'_S}, \mathbf{line_S}, \mathbf{meta_S}$)
From eq. (20)
$R_{GM} \leftarrow R_E \arcsin(\frac{R_E^2+(R_E+h_M)^2-R_M^2}{2R_E(R_E+h_M)})$
From eq. (23)
$R_{GS} \leftarrow R_E \arcsin(\frac{R_E^2+(R_E+h_S)^2-R_S^2}{2R_E(R_E+h_S)})$
From eq. (24) and eq. (25)
$\delta_p \leftarrow \frac{R_{GS}-R_{GM}}{\texttt{sampledPixelSpacing}}$
From eq. (26)
$\Delta \leftarrow \texttt{samplesPerLine}_S - \texttt{samplesPerLine}_M$
From eq. (27)
$\delta_{pC} \leftarrow \delta_p - \Delta$
Compute desired $\mathbf{pixel_S}$
$\mathbf{pixel_S} \leftarrow \mathbf{pixel_M} + \delta_{pC}$
return $\mathbf{pixel_S}, \mathbf{line_S}$

VII. Results and Discussion

Algorithm 1 is applied to E-GLA-D of burst 140 from subproduct no. 5 (see Fig. 8), and it is tracked in the stack of 26 × 2 repeat-pass acquisitions of Fig. 5. Half of this stack is processed using downlinked orbits and the other half utilizes definitive orbits. Note the 1 × 2 image acquired on August 02, 2022, is taken as the main image and the remaining 25 × 2 images acquired from August 10, 2022, to November 22, 2022, are the secondary images.

Algorithm 1 is supplied with the center $(\mathbf{pixel_M}, \mathbf{line_M})$ of E-GLA-D identified in the main image, the corresponding relevant metadata from the main PIF, and the PIF file for each secondary image in the stack. Then, the algorithm calculates the corresponding $(\mathbf{pixel_S}, \mathbf{line_S})$ in each secondary image. It should be noted that $\mathbf{line_S}$ is accurately estimated using Forward/Reverse Geolocation as the state vector perturbations among repeat-pass acquisitions are only impacting the pixel location for the secondary E-GLA-D. Hence, $\mathbf{line_S}$ will not be discussed further. Next, the error in the estimated pixel for each secondary image relative to the truth secondary pixel is calculated as follows:

$$\text{errorPixel}_S = \text{estPixel}_S - \text{truthPixel}_S. \tag{28}$$

Error results from (28) are provided in Fig. 13. It can be noted that for the downlinked orbits $\text{errorPixel}_S \in [1.12, 5.10]$ pixels, while it is $\in [0.12, 2.88]$ pixels for the definitive orbits. Hence, the definitive orbits outperform the downlinked ones. This is expected and it has been previously reported, e.g., see [35]. These results demonstrate the efficacy of the proposed algorithm.

OceanSuite is the software package used for operational ship detection [36], [37]. For ship monitoring in ocean areas of interest such as in DFO's DVD program touched upon in Section I, repeat-pass acquisitions are typically required. Hence, a straightforward approach for allowing OceanSuite to remember past detections and track new ones is presented in Fig. 14 [38]. While a conventional recurrent FA will be detected at the same geographic location in repeat-pass acquisitions, the geographic ground-pixel location for an E-GLA in the repeat-pass secondary image can incur an additional shift either toward or away from the radar, proportional to perturbations in the secondary position state vectors relative to the main ones. In the set-up depicted by Fig. 14, Algorithm 1 is required to estimate the additional pixel shift for the E-GLA. The proposed approach is lightweight and it can be incorporated operationally into OceanSuite with minimal effort.

VIII. Conclusion

With the emergence of a grating lobe in the visible region, the unidirectional planar active phased-array antenna of RCM is effectively rendered into bidirectional antenna with two competing lobes. The electronically steered main lobe of the antenna is pointed at the target of interest while the grating lobe is on the opposite side of the physical boresight. Radar returns from the grating lobe introduce a grating lobe ambiguity. This phenomenon can occur due to sufficiently high electronic steering for the main lobe in either the azimuth or the elevation directions. This work has focused on grating lobe ambiguity due to elevation steering. Using theoretical analysis, simulations, and a real-world dataset, it has been demonstrated in this article that an elevation grating lobe begins to emerge in the visible region of the antenna at an elevation steering angle $\theta_{\text{El}_0} = 17.38°$, which corresponds to a radar look angle $\beta = 50.38°$. Gain for the grating lobe versus loss in the main lobe are proportional to the elevation steering angle.

From a real-world operational perspective, E-GLAs are FA ship detections that are challenging to identify because they resemble legitimate targets. In this article, a novel algorithm is presented for detecting and tracking E-GLAs in repeat-pass RCM imagery. The algorithm takes advantage of the fact that E-GLAs are recurrent in repeat-pass acquisitions. Starting from a center (pixel, line) for a potential E-GLA in a main image, the algorithm finds the corresponding center (pixel, line) in secondary repeat-pass images. If the detection is found to be persistent in two or more repeat-pass acquisitions it can be declared as a verified E-GLA and the proposed algorithm can conveniently track it in subsequent repeat-pass acquisitions.

Because the three RCM satellites are traveling in a 120 m tube, the orbital distance between any two repeat-passes can be up to 240 m which leads to perturbations in the corresponding relative satellite state vectors. This causes the geographic GRD pixel location for E-GLA in the secondary image to shift either toward the radar or away from it, depending on the variations in the secondary state vectors relative to the main ones at the zero-Doppler time where the E-GLA is located in both images. Using the geometry relationships for both the main and secondary images, the proposed algorithm can compensate for this pixel shift. Efficacy for the proposed algorithm is demonstrated through tracking an E-GLA detected by 3MD-based CFAR near Galápagos Islands in a repeat-pass Ship Detection RCM dataset. The RCM stack utilized is comprised of 26×2 repeat-pass images processed using downlinked and definitive orbits, respectively. It is recommended to use definitive orbits when available because they are found to outperform their downlinked counterparts.

Appendix A
Off-Nadir Versus Incidence Angles

In Fig. 12, the off-nadir (or look) angle for the main beam is

$$\beta = 33° + \theta_{\text{El}_0} \tag{29}$$

where 33° is the inclination angle for the RCM antenna [22], and θ_{El_0} is the elevation steering angle. The corresponding incidence angle is given by

$$\theta_i = \beta + \varphi_M \tag{30}$$

where $\varphi_M = (R_{\text{GM}}/R_E)$, as given by (17). Notice in Fig. 12, the 90° complement of θ_i is the grazing angle ψ. For the Ship Detection RCM sub-product in Fig. 7 there are ten beams named SC-28 to SC-37, respectively, that support incidence angles $\theta_i \in [39.82°, 58.24°]$ for the [near, far] ranges. Using (30), the corresponding off-nadir angles $\beta \in [35.87°, 51.11°]$. Note that for beam SC-37 (i.e., subswath no. 10) where E-GLA-D is located, $\theta_i \in [57.07°, 58.24°]$ and the corresponding $\beta \in [50.17°, 51.11°]$, respectively.

Appendix B
R_E at Point B Using WGS-84

In Fig. 12, the geocentric Earth radius R_E at point B is the distance from Earth's center to a point on the spheroid's surface at geodetic latitude where the grating lobe ambiguity is geographically located. In the subsequent equation, the WGS-84 ellipsoid is used where [39].

1) The Earth's equatorial radius, or semi-major axis is

$$a = 6378137.0 \text{ m}. \tag{31}$$

2) The Earth's polar radius, or semi-minor axis

$$b = 6356752.3142 \text{ m}. \tag{32}$$

And for latitude lat in radians, the corresponding Earth radius can be calculated as follows:

$$R_E(\text{lat}) = \sqrt{\frac{\left(a^2\cos(\text{lat})\right)^2 + \left(b^2\sin(\text{lat})\right)^2}{\left(a\cos(\text{lat})\right)^2 + \left(b\sin(\text{lat})\right)^2}}. \tag{33}$$

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From 2010 to 2014, he was a Doctoral Researcher with C-CORE, St. John's, where he developed algorithms for target recognition in SAR imagery. From 2014 to 2017, he was a Senior Engineer with Northern Radar Inc., St. John's, where he co-developed a high-frequency software-defined radar for coastal ocean applications. He also was an instructor for both the graduate and undergraduate Antennas course with the Department of Electrical Engineering, Faculty of Engineering and Applied Science, Memorial University. From mid-2017 to early 2021, he was a Research and Development Scientist with MDA Systems, Richmond, BC, Canada, where he contributed to multiple projects including laser guide stars for optical communications, ground moving target indication and velocity estimation in SAR imagery, and target recognition in ISAR imagery. He is a Professional Engineer (P.Eng.) with the Association of Professional Engineers and Geoscientists of Alberta. Since mid-2021, has joined Defence Research and Development Canada, Ottawa, ON, Canada, as a Defence Scientist, where he has been working on various aspects of SAR and ISAR imagery as well as other related Research and Development topics. Since June 2023, he has been appointed as an Affiliate Faculty with the Fowler School of Engineering, Chapman University, Orange, CA, USA.

Dr. El-Darymli is a fellow of the School of Graduate Studies. During his Ph.D. research work with C-CORE, he was awarded the Ocean Industries Student Research Award offered by the Research and Development Corporation (InnovateNL), Government of Newfoundland and Labrador, for his research project entitled, Synthetic Aperture Radar Detection in Cluttered Environments, and he won the IEEE NECEC 2013 Wally Read Best Student Paper Award. During his M.Sc. work, he developed an award winning Speech to American Sign Language Interpreter System utilizing relevant machine learning techniques.



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Dr. Gierull was elected as a fellow of IET. He initiated and co-organized the first Special Issue on Multi-Channel Space-Based SAR in the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2015. He was a recipient of the Best Annual Paper Award of the Association of German Electrical Engineers in 1998, and a co-recipient of the best paper awards at the International Radar Conference 2004, and the EUSAR 2006, as well as EUSAR 2016. He received DRDC's highest S&T Performance Excellence Award three times, in 2013 and 2020, and again in 2022 in addition to the IEEE Geoscience and Remote Sensing Society 2016 J-STARS Paper Award. He and his team also won the Square Dance Arnold Award in 2021. He has been a Technical Advisor to the Canadian Space Agency on several SAR missions including RADARSAT-2, RCM, and its follow-on. In 2021, the European Space Agency selected him as a Member on its Sentinel-1 Next-Generation Mission Advisory Group. He was an Associate Editor (AE) of *EURASIP's Signal Processing* in 2004. Since 2019, has been serving as an AE for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.